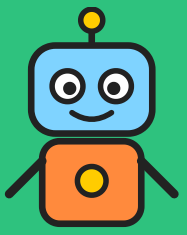
Robotics Project #4 · Kids.JoinDev

Hydraulic Robot Arm

Ages 8+ · 90 min · Build with a grown-up!



🎯 Goal

Use water-filled syringes to lift and grab like a real robot arm.

🧰 You need



stiff cardboard strips



4 syringes (no needles!)



plastic tubing



water with food colouring



paper fasteners (split pins)



tape

🔧 Build it

1. Cut 3 cardboard strips: base, lower arm, upper arm. ☐
2. Join them with split pins so they bend. ☐
3. Fill one syringe with coloured water, connect tubing to an empty one. ☐
4. Tape the empty syringe to the arm so pushing makes it lift. ☐
5. Make a second pair to open and close a cardboard claw. ☐
6. Push the plungers to move. Pick up a paper cup! ☐

⚠️ Safety

Syringes with NO needles only. Work on a tray in case of spills. A grown-up helps with cutting.

💡 How it works

Water can't be squashed. Push one plunger and the water pushes the other one out. Diggers and real robot arms use hydraulics like this.